

Jason Lopez

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Education

University of California, Berkeley – Mechanical Engineering, BS (Aug. 2022 - Dec. 2024)

GPA: 3.54

College of the Canyons – Math, Engineering, and Physics, AS (Aug. 2020 - Jun. 2022)

GPA: 3.82

Skills

CAD & Design: Autodesk Fusion, SolidWorks, Creo, QuickSurface, Peel 3D

Programming: MATLAB, Python, Arduino, LaTeX

Skills: DFM/DFA, 3D Printing, Surface modeling, GD&T, prototype validation, tolerance analysis, manufacturing process selection

Work Experience

Mechanical Design Engineer

Apr. 2025 - Present

Fierro Fab

San Pedro, CA

- Improved fabrication accuracy and repeatability by designing and manufacturing custom dies, bucks, and fixtures, applying DFM principles, tolerance planning, and material selection for sheet metal forming operations.
- Reduced part rework and alignment errors by reverse engineering automotive components using 3D scanning, surface modeling, and precise mounting hole and datum reconstruction, enabling accurate CAD alignment and manufacturable redesigns.
- Delivered production ready CAD models by surface modeling complex automotive panels from mesh data in Autodesk Fusion, incorporating GD&T and manufacturing constraints for CNC machining and formed parts.
- Accelerated prototype turnaround time by operating and managing multiple 3D printers, validating fit, function, and manufacturability prior to fabrication.
- Supported concept to fabrication product development by collaborating with engineers, fabricators, and designers to refine geometry, materials, and manufacturing processes for CNC machining, sheet metal forming, and low volume production.

Mechanical Design Research Intern

Jun. 2024 - Dec. 2024

Additive Manufacturing and Metamaterials Laboratory

Berkeley, CA

- Increased multi-material printer throughput by designing and developing a dual-projector mechanical system and optimizing mechanical layout and assembly.
- Designed modular mechanical components including vats, mounts, connectors, and motion assemblies with attention to tolerances, alignment, and ease of maintenance.
- Created detailed CAD assemblies and technical drawings to support prototype fabrication and iterative testing cycles.
- Improved system reliability by refining component interfaces and structural features based on prototype feedback and testing observations.
- Collaborated with researchers to evaluate material behavior and manufacturability of printed structures, supporting development of conductive printed components.

Instrumentation Engineer Intern

Jun. 2023 - Aug. 2023

Lawrence Berkeley National Laboratory

Berkeley, CA

- Improved installation planning accuracy by deploying and calibrating accelerometers and analyzing vibration data in MATLAB, enabling engineering decisions for sensitive equipment placement.
- Developed data visualization tools and dashboards to communicate vibration trends and experimental results to engineering teams.
- Assisted in experimental setup, wiring, and data acquisition for vibration monitoring systems used in laboratory environments.
- Validated sensor readings and troubleshoot instrumentation issues to ensure reliable data collection during testing.
- Documented testing procedures and results to support repeatability and knowledge transfer across the research group.

Projects

Automatic Chess Board: Developed an automated chessboard capable of moving pieces using a gantry system and electromagnets.

CNC Pen Plotter: Designed and fabricated a belt-driven CNC plotting system including mechanical components and motion control.

Adjustable Laptop Stand: Designed and prototyped a load-bearing adjustable stand optimized for manufacturability and assembly.

Delay Guitar Pedal: Engineered and assembled a custom delay guitar pedal by designing a custom PCB in KiCad, integrating electronic components and controls, and designing 3D-printed housing.